Zheye Song

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Education

University of North Carolina at Chapel Hill — *Aug 2026 – Present*
M.S. in Biostatistics

Fudan University, 87/100 — *Sep 2022 – Jun 2026*
B.S. in Mathematics and Applied Mathematics

The Hong Kong Polytechnic University, 3.97/4.3 — *Sep 2024 – Jan 2025*
Exchange Student

Research Projects

Causal Inference in Metric Spaces via Distance Profiles — *Aug 2025 – May 2026*
Advisor: Zhiheng Zhang, Tenure-Track Assistant Professor, SUFE — *Under Review*

- Developed a distance-profile causal inference framework for object-valued outcomes by transforming metric-ball events in separable metric spaces into geometry-indexed binary potential-outcome contrasts.
- Proved pointwise identification, doubly robust/AIPW estimation properties, Neyman orthogonality, and auxiliary-stabilized finite-grid simultaneous inference under causal identification and high-level nuisance conditions.
- Validated the framework across Euclidean, Wasserstein, SPD, graph, and tree-valued outcomes, with semi-synthetic connectome and network benchmarks supporting calibrated coverage, global testing, and robustness diagnostics.

Relative-Belief Multiple Testing under General Additive Loss — *Jan 2026 – Present*
Collaborator: Qiaoyu Liang, Ph.D. student, University of Toronto — *Ongoing*

- Building on "An Evidential Route to Asymptotic Bayes Optimality under Sparsity" (previous work), extended the relative-belief ABOS framework from additive 0-1 loss to general additive loss by deriving the loss-adjusted Bayes oracle, fixed-threshold ABOS window, and risk scale.
- Developed a regular scale-prior relative-belief threshold theory using local prior-predictive expansion, yielding ABOS conditions for one-group working priors.
- Unified normal and uniform relative-belief rules under the regular scale-prior framework and organized additional regular-prior cases including Laplace, Cauchy, and Student-t priors.

Survey-Based Health Risk Modeling with Interpretable Machine Learning — *Nov 2025 – Present*
Advisor: Kun Tang, Associate Professor, Tsinghua University — *Under Review*

- Participated in questionnaire data cleaning and feature preparation for a large-scale student health survey, producing analysis-ready inputs for downstream modeling.
- Built and ran the full machine-learning pipeline, including imputation, stratified train/test evaluation, six-model benchmarking, class-imbalance handling, and interpretable model analysis with feature importance and SHAP outputs.

Modular Causal Meta-Learners for CATE Estimation — *Apr 2025 – Apr 2026*
Research Project

- Built a BART-based cross-fitted nuisance layer and modular CATE estimator library, separating S-, X-, R-, and BCF-inspired correction principles under a common implementation.
- Implemented HybridFull, an overlap-aware orthogonal-residual stacking rule that learns simplex weights over candidate CATE estimators within propensity strata, and evaluated it across S1–S6 simulations plus IHDP/JOBS benchmarks.

Technical Skills

Programming: Python, R, Git, LaTeX

Honors and Awards

Freshman Scholarship, First Prize — *2022*
Fudan University

Basic Discipline Scholarship — *2023–2024*
Fudan University